

SCHMIDT CAPITAL

An autonomous options-trading agent with deterministic risk gates

Alpaca AI Trading Agents Hackathon · lablab.ai × Alpaca · Sep 2026 · solo entry

Paper account PA3KGE66JM72 · github.com/Welpphantom/alpaca-trading-challenge

Sell time, define risk, survive anything

- **Core sleeve:** 0–5 DTE credit spreads on XSP at $\sim 15\Delta$ ($\sim 1\sigma$) — the variance risk premium, realized daily by theta
- **Every position has a known max loss at entry.** No naked anything, ever
- **Rule-based exits:** 60% profit-take · debounced 2×-credit stop · expiry handling
- **Expiry ladder + delta-scaled credit floor:** a missing 15Δ trade becomes NO trade, never a 30Δ trade
- **Why it fits one week:** time passing is the only edge guaranteed to show up during the competition

The LLM proposes, code disposes

RESEARCH STRATEGY RISK (veto power) EXECUTION

- Claude reads the market through Alpaca's official MCP server — snapshots, news, movers, chains
- Order-placing MCP tools: whitelisted out AND explicitly banned — it physically cannot trade
- Tighten-only, enforced in code: every recommendation is min()-ed against config — never looser
- Day-one proof: caught the wrong event on our static calendar (Warsh at Jackson Hole)
- Every decision journaled with written rationale — visible on the live dashboard

Eight gates, zero trust, no panic

- **Kill switches:** daily -2% (blocks new risk) · competition -5% (flatten + freeze)
- **Sizing gates:** \$1,500/trade · \$10,000 open-risk budget · shrink-before-veto
- **Sustained-breach debounce:** no single reading — glitched data pauses the clock, it never triggers action
- **Exits are re-pegged limit orders** — never market orders into an options book
- **Live:** rode a $-\$94$ dip back to green without panicking; stopped two positions at $\sim 1-2\times$ credit in a selloff
- **Wind-down protocol:** book fully flattened and realized one hour before the judging deadline

Everything verified live before it mattered

- **Trading API:** atomic MLEG spreads — signed net pricing, close intents, BP = defined risk (verified live)
- **MCP server:** the research agent's entire market-reading surface
- **CLI:** independent equity cross-check in every end-of-day report
- **Paper env:** two full shakedown sessions before kickoff; restart-safe, blackout-aware, self-launching
- **Live dashboard:** equity curve, position book, and the full decision journal — Streamlit

equity \$100,018 (green) · 10 trades, all realized · 70% win rate · max drawdown ~\$139 · flattened itself before the c